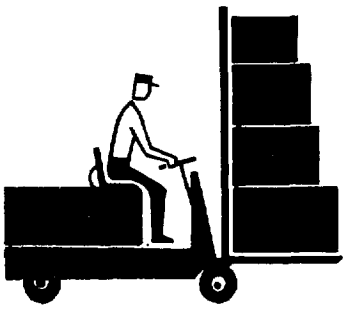
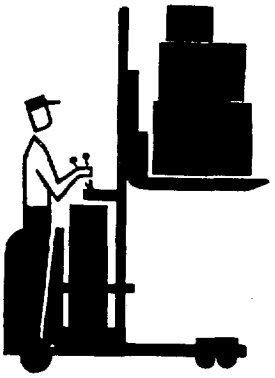
OPERATING & MAINTENANCE

INSTRUCTIONS

EV-1* SCR CONTROL for ELECTRIC VEHICLES





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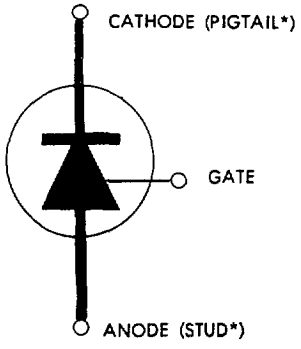
The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

GENERAL ELECTRIC

WHAT IS AN SCR?

Since the heart of the control is a silicon controlled rectifier (SCR), a general understanding of the characteristics of the device will be helpful. The SCR is a semi-conductor rectifier used as a latching switch; i.e., it may assume either a conducting or nonconducting state (On or Off).



The SCR can be turned on by a momentary application of control current to the gate. To turn it off, it is necessary in addition to removing the turn-on signal from the gate, either to remove all power from the SCR or to apply momentary reverse voltage between cathode and anode.

* Typical of SCR as used in GE control for electric vehicles.

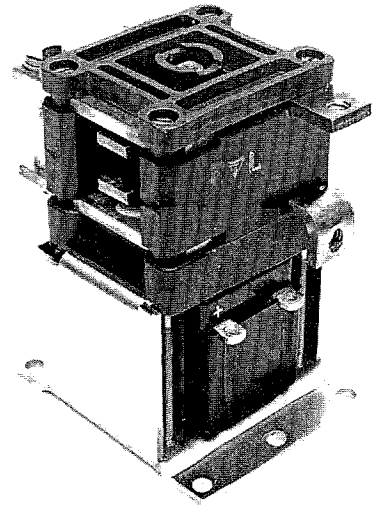


Fig. 2. Typical contactor

PHOTOS OF CONTROL

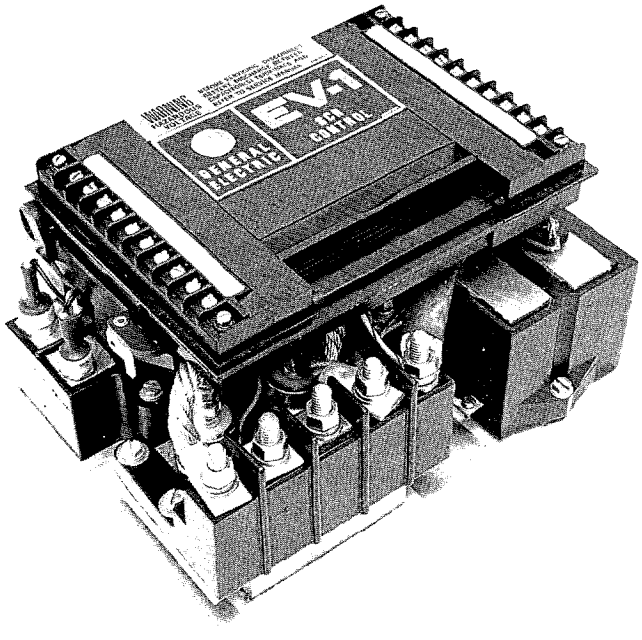


Fig. 1. Typical SCR static panel

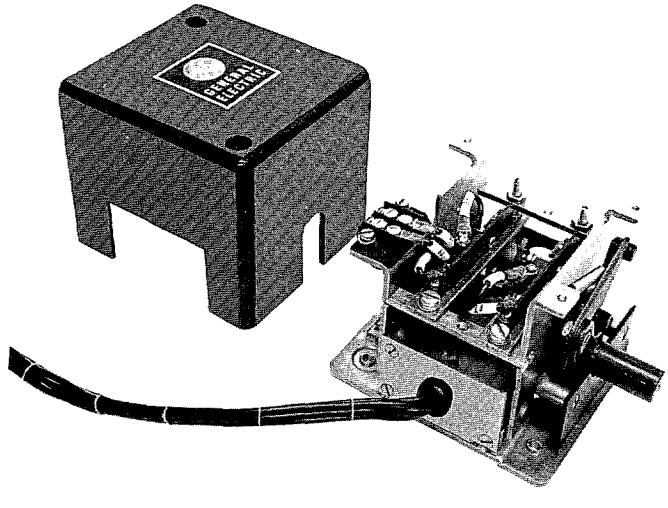
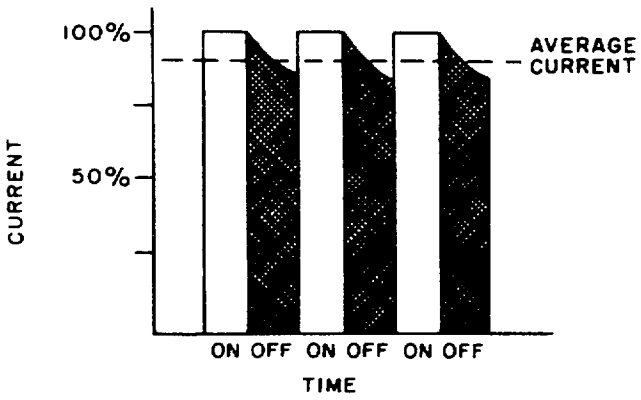


Fig. 3. Typical accelerator switch with cover removed

rent and the inductive flyback current. It should be noted that the average motor current measured will be greater than the average battery current. The SCR control, in effect, converts battery current at battery volts into a higher motor current and a lower motor volts.

The time for the next cycle to start is determined by the time that the control card takes to oscillate. This frequency of oscillation is controlled by the potentiometer in the accelerator. Slow speed is obtained by having maximum ohms in the potentiometer. As the resistance in the potentiometer decreases, the speed of the motor increases. With level operation, the SCR circuit is capable of delivering approximately 85- to 90-percent speed. For full-speed operation, the 1A contactor is closed to apply full battery voltage across the motor.



CONTROL FEATURES

- OSCILLATOR — The oscillator section of the card has two adjustable features and one fixed feature. With the accelerator potentiometer at maximum ohms, the creep speed can be adjusted with a trimpot on the card. Top speed is fixed by card design, and is obtained with the accelerator potentiometer at minimum ohms. The control will provide 85 to 90 percent of available voltage. Fig. 7 shows the oscillator frequency curve. The controlled acceleration (C/A) trimpot can be set to provide different rates of acceleration with the accelerator potentiometer at minimum ohms. This feature provides two acceleration rates — one at start up and another on re-acceleration after plug.

- CURRENT LIMIT — This circuit monitors motor current by utilizing a sensor in series with the armature. The information detected across the sensor is fed back to the card so current may be limited to a preset value. If heavy load currents are detected, this circuit limits the average battery current to a value based on the maximum thermal and commutation rating of 1 REC. Because of the flyback current through 3 REC, the motor current usually runs 2 to 3 times the battery current-limit value. The current limit is set with the (CL) trimpot.

NOTE: On some truck models a jumper is supplied between R1 and R2 on the oscillator card. In these cases, refer to the truck manufacturer's instruction book.

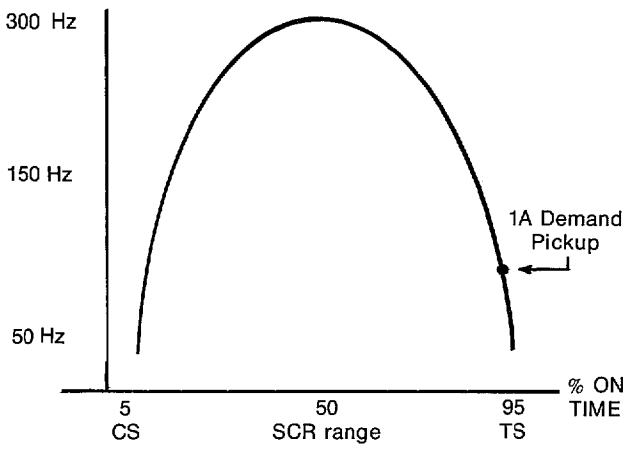


Fig. 7. Oscillator frequency curve

- PLUGGING — Slow down is accomplished when reversing by providing a small amount of retarding torque for deceleration. If the vehicle is moving and the directional lever is moved from forward to reverse, the motor field is reversed. During the 1 REC off times the motor armature, driven by the inertia of the vehicle, acts as a generator. This generated current passes through 4 REC and the sensor. The plug signal is initiated by the fact that the directional switch has moved from one direction to the other. The oscillator section assumes a plug current limit mode and regulates the armature current to obtain the desired braking torque. During the plugging mode the control oscillates at about 10 hertz resulting in a smooth plug. The distance of the stopping can be set with the (plug) trimpot.

- ANTI-ROLLBACK — This non-adjustable feature provides full SCR torque to restart a vehicle on an incline. The memory for this function is the directional switch. When stopping on an incline, the directional switch must be left in the original or off position to allow the control to assume full power when restarted.

CONTROL FEATURES (Cont'd)

- PULSE MONITOR TRIP — This feature is designed to monitor the control for 1A welds, 1 REC shorted or 1 REC failure to commutate. If 1 REC is shorted or 1A is welded, PMT will test for this

TABLE 3

MISOPERATION OF OTHER FEATURES

SYMPTOM	PROBABLE CAUSE
3A. 1A or FW contactors close with key switch.	• Check drivers for short from terminals 2 to 3 by disconnecting wires to terminal 1 on the driver. (4E) • Replace control card. (4A)
3B. 1A will not close at run (percent pickup).	• Check connection from control card R8 to driver terminal 1. • Check coil continuity. • Jumper 1A driver terminal 2-3 and close key switch to operate 1A coil. • Jumper a 4.7k resistor (2-watt) from positive to card terminal R8, with key switch closed 1A should close. • Check that motor volts (T2 to Neg) are more than 85% of battery volts. If not, control may be in thermal cutback. • Replace control card. (4A)
3C. 1A will not close at SCR stall (time pickup). (Check truck diagram to see if 1A switch closes card circuit R4 to R6.)	• Check 1A switch circuit between card terminals R4-R6. • Measure volts across TP. If volts are above 0.6, control is in thermal cutback. Allow to cool, and recheck 1A function. • Replace control card. (4A)
3D. FW contactor will not close after 1A pickup.	• Check PU setting on card. • Check card terminal R6. After 1A closing this point should be 2 volts or less. • Check 1A switch. • Check FW driver. (4E). Close key switch and jumper a 4.7k resistor from card terminal L3 to card terminal R9. Coil should operate.

3E. FW contactor will not drop out with increasing load.	• Check dropout setting on card. • Replace control card. (4A)
3F. Stiff plug. Severe reversal.	• Check plug adjustment setting on card. • Check 4 REC for open circuit. (4H) • Replace control card. (4A)
3G. Very soft reversal.	• Check plug adjustment setting on card. • Replace control card. (4A)
3H. Blown power fuse. Very hot power cables.	• Check 3 REC for short. (4H) (Possible damage also to 1 REC and transformer module.)
3J. Hourmeter feeder faults:	
(1) Pump contactor closes when either F or R direction is selected.	• Diode shorted 3 to 4. (4H) Replace hourmeter block.
(2) One direction okay; opposite direction picks up *both* F and R.	• Diode shorted 1 to 4 *or* 2 to 4. (4H) Replace hourmeter block
(3) Either direction selected picks up *both* F and R.	• Diode shorted 1 to 4 *and* 2 to 4. (4H) Replace hourmeter block.

TABLE 4

CHECKING COMPONENTS

4A. Main SCR Control Card

All trouble-shooting is written to check all outside devices and eliminate them as the source of symptoms. The conclusion being then that the card is faulty.

1. Instructions for Removal of Card
   a. Remove the four (4) screws shown in Fig. 9.
   b. Jack out the right- and left-hand terminal board, using a screwdriver in the slots, (leaving the wires intact) as shown in Fig. 10.
   c. Pry open the latches carefully with a screwdriver as shown in Fig. 11.
   d. Jack out the bottom plug with a screwdriver as shown in Fig. 12.

The card can be removed by hinging 10 degrees and pulling out, or, if panel components (not related to card hinge mountings) are to be replaced, disregard all instructions above except "C" and the card will hinge up to 90 degrees.

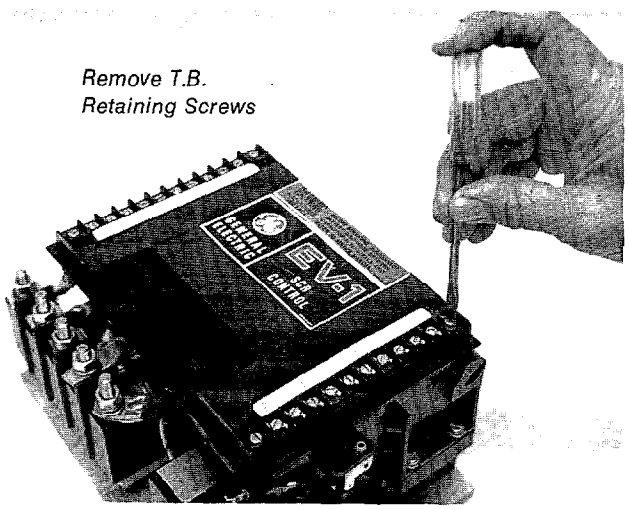


Fig. 9.

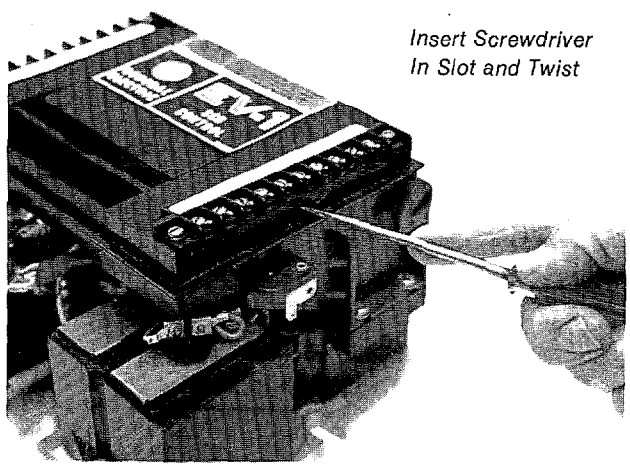


Fig. 10.

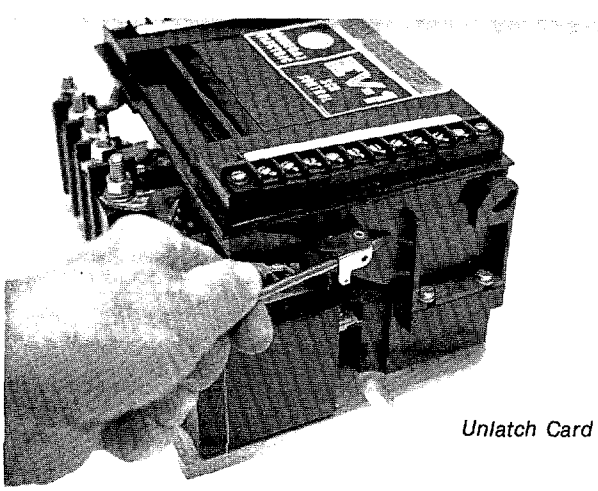


Fig. 11.

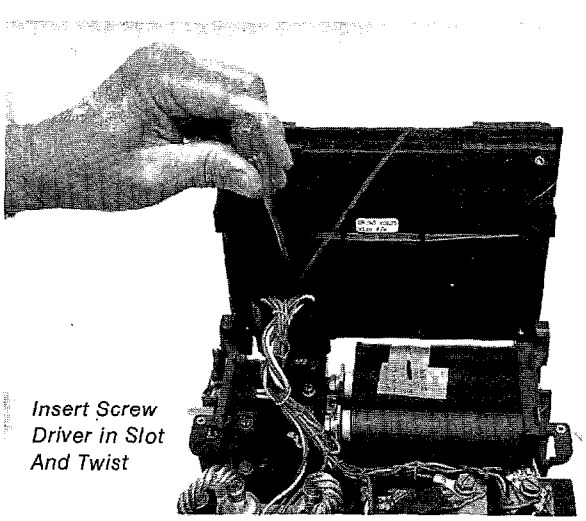


Fig. 12.

4B. Capacitor 1C

Disconnect battery and discharge capacitor. Measure ohms through the capacitor using the R x 10,000 scale. Meter should read zero and then swing to above 100,000 ohms. Replace capacitor if above reading is not obtained.

4C. Contactors F, R, 1A, and P

75-ampere contactors (see GEH-3099)
150-ampere contactors (see GEH-4469)
300-ampere contactors (see GEH-4469)

NOTE 1. Control is arranged so that F and R do not break current. Check to see that 1A drops out ahead of F or R.
NOTE 2. F and R contactor coils are polarity sensitive. The left-hand terminal must be connected to positive.
NOTE 3. All 72 volt and above coils are polarity sensitive. The left-hand terminal must be connected to positive.

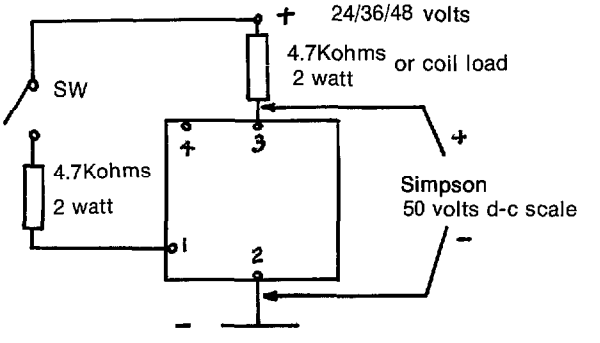
4D. Potentiometer in Accelerator

To check operation of the potentiometer, disconnect battery and disconnect wires at card terminal R4 and R5. Connect a VOM to wire removed with scale set to R x 100. With accelerator in creep speed position, the ohms reading should be 4800 to 6000 ohms. With accelerator in top speed position, reading should be 200 ohms or less. With wire disconnected as above, check for resistance of 1 megohm or higher from pot wires to truck frame.

4E. Driver Module

(IC 3645CPM1RDA2 and IC3645CPMIRDB2)

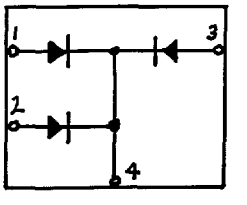
(a) Connect circuit as shown.
(b) Voltmeter should read battery volts with switch open.
(c) Close switch and meter reading should be 2 volts or less.
(d) Move load resistor to terminal 4 and repeat steps (b) and (c).



NOTE: For 72 volt, use 8.2 Kohms 2-watt resistors.

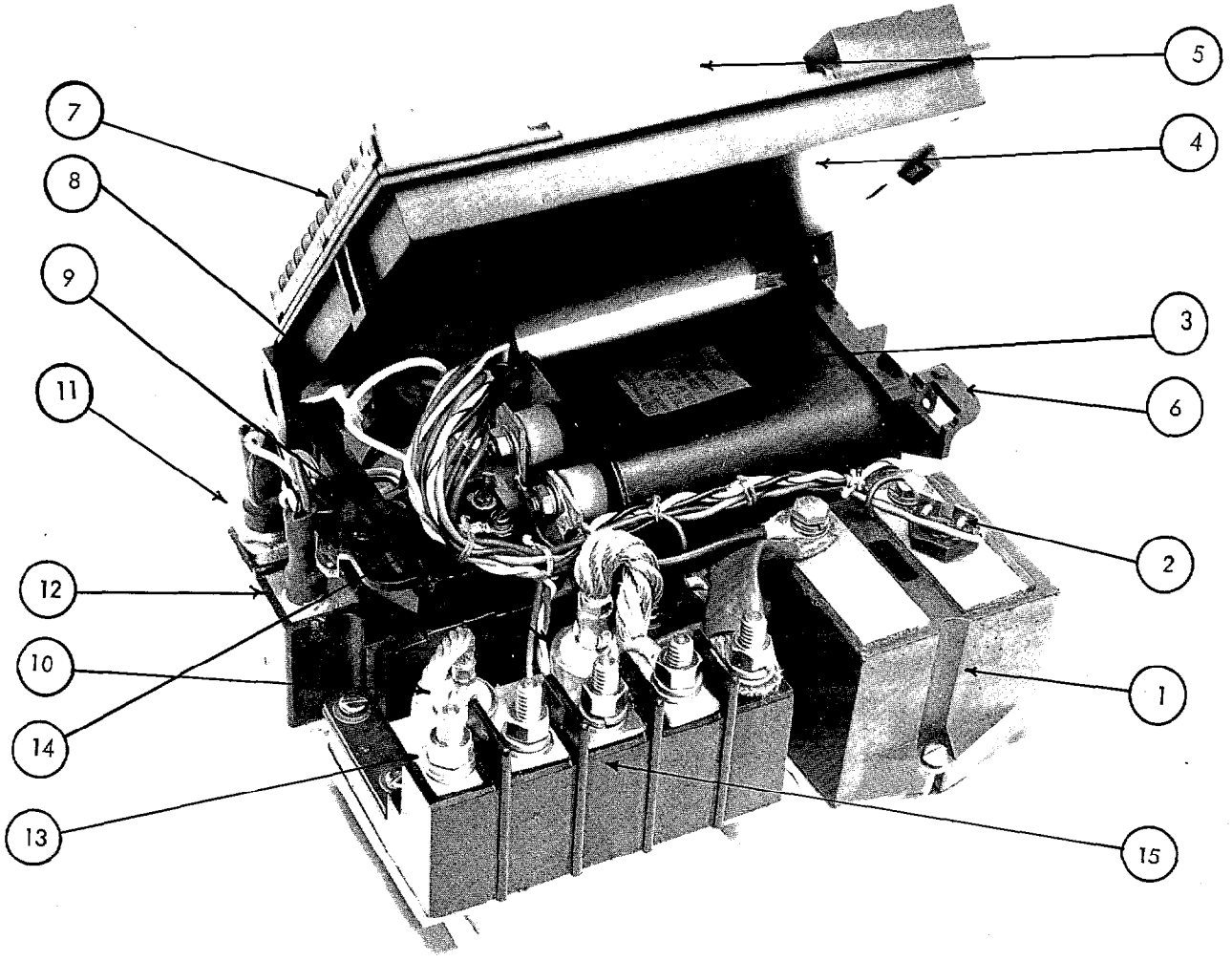
4F. Hourmeter Module

Check individual diode circuits with trouble light or Simpson. (4H)



TYPICAL PHYSICAL ARRANGEMENT AND IDENTIFICATION OF COMPONENTS

(Refer to wiring diagram furnished with truck for precise arrangement of components.)



(1) Main SCR (1 REC)
(2) Thermal Protector
(3) Commutating Capacitor
(4) Oscillator Card
(5) Card Adjustments
(6) Quick Card Release
(7) Card Connection Block
(8) Card Connector
(9) Flyback Diode (3 REC)
(10) Plugging Diode (4 REC)
(11) Turn-off SCR (2 REC)
(12) Charging SCR (5 REC)
(13) Power Connections
(14) Filters for 2 and 5 REC
(15) Motor Current Sensor
(Located behind middle power connector)

Transformer and choke (1X) located in encapsulated block under capacitor. 3 REC filter (23 FIL) located under pigtail of the diode.

Fig. 14. Typical EV-1 SCR panel*

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